MIDTERM TEST, FALL 2025 **Course: CS365 - Intro. to COA** **Duration:** 60 **minutes**	**Student name:** .... **Student Id:** .... **Order:** ...... **Class Id:** ....

Assessors' name and signature	**Examiners' name and signature**	**Total score**

Test Id: 1 **(This test consists of 10 questions)**

Note: *No electronic devices (laptops, smart devices, etc except calculators) are allowed.*

Questions with one correct answer

Question 1. What is the decimal value of this 8-bit two's complement number 1100 0101 after shifting the number right two bits?

- ☐ 15
- ☐ -49
- ☐ 49
- ☐ -15

Question 2. Suppose that register `$t2` contains `0x00000D00` and register `$t1` contains `0x00003C00`. Which of the following MIPS instructions places the value `0x00000C00` in register `$t0`?

- ☐ `and $t1, $t2, $t0`
- ☐ `add $t1, $t2, $t0`
- ☐ `and $t0, $t1, $t2`
- ☐ `add $t0, $t1, $t2`

Question 3. Which of the following MIPS codes loads the 32-bit constant `0x00300100` into register `$s0`?

- ☐
```
lui $s0, 256
addi $s0,$s0, 48
```
- ☐
```
lui $s0, 46
addi $s0,$s0, 256
```
- ☐
```
lui $s0, 48
addi $s0,$s0, 256
```
- ☐
```
lw $s0, 256
add $s0,$s0, 48
```

Question 4. What decimal number is represented by the following IEEE 754 single precision float?

1100 0001 0101 0110 0000 0000 0000 0000

- ☐ 12.375
- ☐ -12.375
- ☐ 13.375
- ☐ -13.375

Question 5. What is the Verilog code that creates a 32-bit value with the pattern 0101...01?

- ☐ `{2{16'b01}}`
- ☐ `{16{2'b01}}`
- ☐ `{8'b1,4'b0}`
- ☐ `32'b01`

Question 6. Suppose that we have two computers A and B implementing the same instruction set architecture. Computer A has a clock cycle time of 1 ns and a CPI of 2.0 for a program P, and computer B has a clock cycle time of 1.5 ns and a CPI of 1.2 for P. How much faster is computer A than computer B for this program?

- ☐ 1.5 times
- ☐ 1.3 times
- ☐ 1.2 times
- ☐ 0.9 times

Constructed-response questions

Question 7. Assume that \$a0 initially contains 5 and \$v0 is used for the output. Add comments to the following MIPS code and describe in one sentence what it computes.

```
begin:  addi $t0, $zero,0
        addi $t1, $zero,1
loop:   slt $t2, $a0, $t1
        bne $t2, $zero,fin
        add $t0, $t0, $t1
        addi $t1 $t1, 2
        j loop
fin:    add $v0, $t0, $zero
```

Question 8. What is the MIPS assembly code corresponding to the following machine code?

```
10001101001010000000010010110000
00000001001001000010000000100000
10101101001010000000010010110000
```

Question 9. Compile the following C program into MIPS assembly code.

```
int func (int n)
  {
    int i;
    for (i=0;i<n;i=i+1)
      if (i==n) return i;
  }
```

Question 10. Construct the full truth table described by the following Verilog module.

```
module func {A, B, C, S, D};
   input A, B, C;
   output S, D;
   assign S=A^B^C;
   assign D=(A&B)|(B&C)|(C&A);
endmodule
```

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<table>
<tr><td colspan="2">MIDTERM TEST, FALL 2025
Course: CS365 - Intro. to COA
Duration: 60 minutes</td><td>Student name: ....
Student Id: .... Order: ......
Class Id: ....</td></tr>
<tr><td>Assessors' name and signature</td><td>Examiners' name and signature</td><td>Total score</td></tr>
</table>

Test Id: 2 **(This test consists of 10 questions)**

Note: *No electronic devices (laptops, smart devices, etc except calculators) are allowed.*

Questions with one correct answer

Question 1. What is the decimal value of this 8-bit two's complement number 1110 0101 after shifting the number left two bits?

☐ −148 ☐ 108 ☐ 148 ☐ −108

Question 2. Suppose that register `$t2` contains `0x00000D00` and register `$t1` contains `0x00003C00`. Which of the following MIPS instructions places the value `0x00004900` in register `$t0`?

☐ `and $t1, $t2, $t0` ☐ `add $t1, $t2, $t0`

☐ `and $t0, $t1, $t2` ☐ `add $t0, $t1, $t2`

Question 3. Which of the following MIPS codes loads the 32-bit constant `0x01000030` into register `$s0`?

☐
```
lui $s0, 256
addi $s0,$s0, 48
```
☐
```
lui $s0, 46
addi $s0,$s0, 256
```
☐
```
lui $s0, 48
addi $s0,$s0, 256
```
☐
```
lw $s0, 256
add $s0,$s0, 48
```

Question 4. What decimal number is represented by the following IEEE 754 single precision float?

1011 1111 0100 0000 0000 0000 0000 0000

☐ −0.375 ☐ −1.75 ☐ 0.375 ☐ −0.75

Question 5. What is the Verilog code that creates a 32-bit value with the pattern 0101 . . . 01?

☐ `{2{16'b01}}` ☐ `{2{16'h5555}}`

☐ `{16'b01,16'b01}` ☐ `32'b01`

Question 6. Suppose that we have two computers A and B implementing the same instruction set architecture. Computer A has a clock cycle time of 2 ns and a CPI of 1.5 for a program P, and computer B has a clock cycle time of 1.5 ns and a CPI of 1.2 for P. How much faster is computer A than computer B for this program?

☐ 0.9 times ☐ 1.3 times ☐ 1.2 times ☐ 0.6 times

Constructed-response questions

Question 7. Assume that \$a0 initially contains 7 and \$v0 is used for the output. Add comments to the following MIPS code and describe in one sentence what it computes.

```
begin:  addi $t0, $zero,0
        addi $t1, $zero,2
loop:   slt $t2, $a0, $t1
        bne $t2, $zero,fin
        add $t0, $t0, $t1
        addi $t1 $t1, 1
        j loop
fin:    add $v0, $t0, $zero
```

Question 8. What is the MIPS assembly code corresponding to the following machine code?

```
10001101001010000000010010110000
00000010010010000100000000100000
10101101001010000000010010110000
```

Question 9. Compile the following C program into MIPS assembly code.

```
int func (int n)
  {
    int i;
    while (i<n)
      {i=i+1;} return i;
  }
```

Question 10. Construct the full truth table described by the following Verilog module.

```
module func {A, B, C, S, D};
   input A, B, C;
   output S, D;
   assign S=A&B&C;
   assign D=(A^B)|(B^C)|(C^A);
endmodule
```